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Project Title:	Bioreactor Technology Transfer		
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Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biotechnology		
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Space Biology Special Category:	None		
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Zip Code:	20892-1855	**Congressional District:**	8
Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
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Task Description:	The NASA-NIH Center for Three-Dimensional Tissue Culture supports a number of scientists who work on projects related mainly to cell-cell interactions, cell differentiation, and the pathogenesis of infectious diseases. Many mainstream medical research studies have important implications for human space exploration and the Center helps promote the transfer of technology and interests between NASA and NIH. Intramural and extramural scientists launch pilot projects involving the RWV Bioreactor as extensions of their main research projects. Based on preliminary results, a research group can apply for a small grant proposal for a study involving the RWV. Additionally, many in-house projects are ongoing at the Center. Below are the current projects at the NASA/NIH Center: 1) understanding the role of microgravity on the immune function of lymphoid tissue and cells, 2) studying the role of extracellular matrix and growth factors on differentiation of embryonic stem cells, 3) isolating and characterizing breast cancer stem cells, and 4) attempting to propogate Noroviruses in vitro. The goal of the Center is to maintain a core facility at NIH for use of the NASA Rotating Wall Vessel (RWV) Bioreactor and related three-dimensional tissue culture systems for studying the basic mechanisms of normal and pathological intercellular interactions in multi-cellular systems, and how these interactions can be influenced by modeled microgravity. Investigators provide biological material or instruct the Center staff on experiment design, dissection and culture techniques. The investigator provides mentorship and is available for questions beyond staff expertise. The Center researchers prepare and culture the cells or tissues, and perform the primary analysis, including gathering data on tissue metabolism and tissue cytoarchitecture. The Center's staff also provides services such as immunocytochemistry, ELISA, confocal microscopy, electron microscopy, and flow cytometry. The primary investigator performs specialized analyses specific to their project. Successful pilot projects can then apply for funding, which keeps the staff of the Center available for new pilot projects, and promotes technology transfer by recruiting scientists to work on three-dimensional tissue culture full time. Because of our NASA funding, and because of the possibility that results obtained in suspension culture may relate to health problems of astronauts, or present unique opportunities for studying physiology in microgravity, there are possibilities of flight experiments for PIs of NASA/NIH projects. Currently we have flown one cellular experiment on the International Space Station (ISS), and we are currently running fluid dynamics experiments on the ISS. The Center provides researchers an opportunity to develop new model systems for biomedical research and provides access to the RWV, which allows cellular growth, aggregation, and development of in vivo-like structures. This vessel allows the culture of cells under minimal shear forces in a well-oxygenated medium under conditions that mimic microgravity. RWVs and experienced staff are available for testing tissues, primary cell cultures, and cell lines under these conditions that facilitate cell-cell interactions and promote differentiation. NASA funding for equipment has enabled us to procure state-of-the-art microscopes to determine structure and function relationships between cells of developed tissues. The design of the Center allows investigators to assess potential benefits of the RWV for their specific aims without having to commit major resources, which greatly increases the number of investigators who will explore the RWV as a tissue culture opportunity and apply their knowledge and experience to space research. Results from projects at the Center benefit space research, since the RWV models some aspects of microgravity, but since they stem from basic medical research, they have many terrestrial benefits as well.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Since most of the studies conducted at the Center stem from mainstream medical research, the results obtained benefit not only space research, but the entire medical research community. The project on understanding the role of microgravity on the immune function of human lymphoid tissue and lymphocytes has important implications for ground based research. Potential terrestrial benefits of these experiments include an understanding of the nature of microgravity-triggered defects in lymphoid tissue responses that may uncover new pathways in the cellular and molecular regulation of the human immune system, as well as use of the ground-based system to test lymphoid tissue susceptibility to certain pathogens and latent viruses, and to test countermeasures or therapies for infections. Studies on differentiation of embryonic stem cells have far reaching earth benefits. Results from these studies are critical for understanding the factors that drive stem cells into various cell lineages, which is important for the whole medical community. Studies on breast cancer stem cells have important implications for early detection and treatment of breast cancer. Normal mammary epithelial stem cells represent potential targets for tumorigenesis and understanding the events that occur when normal or high-risk breast stem cells progress to cancer will provide a more comprehensive insight into early etiologic and molecular events in breast neoplasia and provide unique markers for imaging and targeted therapy. Studies on noroviruses also have earth benefits. Major limitations in the study of the noroviruses have been the failure to propagate them in cell culture. The ability to grow noroviruses in cell culture would be a major advance in the study of these human pathogens.
	To support human exploration of space, it is essential that we assure that the highest standards of medical care be available to astronauts, even during space flights of extended duration. Towards that aim, it has been invaluable to the mission of space exploration to have a Center that promotes the exchange of technology and interests between NASA and NIH. In 2007, intra-mural scientists worked at the NASA/NIH Center for Tissue Culture on four various projects related to the effects of microgravity and modeled microgravity (in the NASA Rotating Wall Vessel Bioreactor (RWV)) on both physiology and pathophysiology of human cells and tissues, as well as developing the means for studying the behavior of various types of stem cells. During 2007, the Center scientists have continued efforts to study immune dysfunction in microgravity, including optimizing conditions for flight experiments onboard International Space Station (ISS), studying role of extracellular signals on differentiation of embryonic stem cells, and isolating and characterizing breast cancer stem cells using three-dimensional tissue culture conditions. In addition a former study was revived to develop a tissue culture model for replication of noroviruses. 1. Study of immune dysfunction of human lymphoid tissue in microgravity. We have found in previous studies that human lymphoid tissues and cells cultured in RWVs and onboard the ISS remain metabolically active, however, they demonstrate impaired antibody and cytokine responses to recall antigens

Task Progress:	and polyclonal activators. We found that cells activated in microgravity (modeled or true) did not increase glucose consumption, increase antibody production, or increase cytokine/chemokine secretion in response to polyclonal challenge compared to their static ground-based counterparts. However, if cells were challenged with activators prior to exposure to either modeled or true microgravity their glucose consumption and antibody and cytokine responses did increase. The consistent results between modeled and true microgravity suggest that RWVs provide a useful model of some aspects of microgravity. Current studies are now focused on understanding the mechanisms of immune dysfunction in ground-based experiments with the RWV, looking at the roles of both T and B cells in the immune dysfunction. We are using various flow cytometry assays to track activation and differentiation of T and B cells in ex vivo tonsil cultures and cell culture suspensions of lymphocytes, as well as evaluating antibody and cytokine/chemokine/growth factor secretion using multiplexed bead assay for analysis of cytokines, chemokines, and growth factors. Previous experiments on board the International Space Station (ISS) have also helped to optimize culture conditions within the TCM (tissue culture modeule) culture system to ensure better science with future experiments. We were able to develop better mixing and bubble removal procedures in microgravity as well as promote interactive science between the flight crew and the ground team. Continued studies will help to develop a complete protocol for future tissue culture investigations in ISS. 2. Effects of extracellular signals on the differentiation of embryonic stem cells (ESC) in three-dimensional static and dynamic culture systems. Our studies have shown that both soluble factors and the type of ECM seem to be critical in directing differentiation and the formation of tissue-like structures. In addition, the composition of the ECM as well as the physical properties (rigidity, etc) of the ECM may influence cell morphology, proliferation, and differentiation. The cell shape acquired by the ESCs, which is mediated by the cell-cell and cell-matrix interactions, may be related to proliferation and lineage specific differentiation. We were able to stimulate ESC into cardiomyocyte lineages without addition of any growth factors and future studies are planned to further enhance this differentiation. New studies were directed toward differentiation into lymphoid lineages using conditioned medium from human tonsil cultures, stromal cells derived from tonsillar tissue, and decellularized tissue. Preliminary experiments did not demonstrate any advanced differentiation into lymphoid cells but further analysis is concentrating on earlier differentiation stages such as CD34. We hope to couple these differentiation studies with HIV experiments. To use stem cells in an HIV therapy application, it would be important to know the susceptibility to HIV infection of the embryonic stem cell, and at which stage of differentiation is it most and least susceptible. The HIV receptors and co-receptors being expressed by the stem cell at those stages, as well as production of certain HIV-inhibitory chemokines, could be important weapons against HIV infection and exploiting the function of those proteins could be used in therapy. 3. Isolation and Characterization of Breast Cancer Stem Cells Using Three-Dimensional Tissue Culture Conditions. Experiments/Results: We developed protocols for culturing human breast tissue in microgravity conditions. Tissue blocks were inoculated into RWV bioreactor in culture medium. Tissue samples were collected during the culture experiment so that viability and proliferation activity of epithelial cells could be examined. Our results demonstrated that cultured tissues blocks showed high viability and potential proliferation of cells up to one month after the initiation of the culture, documented by mitotic figure density. To test for proliferation, we labeled samples for the proliferation marker Ki-67. To confirm the epithelial origin of proliferating cells, samples were labeled for epithelial (wide spectrum cytokeratin antibody) markers. Together these results demonstrated high proliferative activity of the epithelial cells surrounding ducts. In conclusion our data indicate that culturing of human breast tissues under microgravity increase fraction of epithelial cells with high proliferative activity. Thus there is a possibility that the origin of this proliferation is stem cells. 4. Growth of Noroviruses in the RWV Bioreactor. Previous studies with cultures of INT-407 cells inoculated with a stool filtrate containing Norwalk virus, a Genogroup I (GI) norovirus showed suggestions of cytopathic effect on cells and we detected viral RNA by RT-PCR for up to two weeks post-inoculation. A new report using this cell line reported infection using a pool of stool filtrates was used as the inoculum (containing one GI and two GII norovirus strains). We decided to revisit the INT-407 cell line in order to confirm this new report with our improved detection methods. INT-407 cells were cultured for several weeks on microcarrier beads and on a new mucosal substrate in bioreactors and then inoculated with Norwalk virus (GI) and infection will be determined by quantitative RT-PCR. In addition small and large intestine biopsies were placed in three-dimensional culture and infected with the GI virus and evaluated for infection. We are also investigating whether ex vivo tonsil cultures support norovirus infection, as others groups have reported infection of macrophages with feline caliciviruses.
Bibliography Type:	Description: (Last Updated: 10/09/2012)
Articles in Peer-reviewed Journals	van Laar JM, Melchers M, Teng YK, van der Zouwen B, Mohammadi R, Fischer R, Margolis L, Fitzgerald W, Grivel JC, Breedveld FC, Lipsky PE, Grammer AC. "Sustained secretion of immunoglobulin by long-lived human tonsil plasma cells." Am J Pathol. 2007 Sep;171(3):917-27. PMID: 17690187 , Sep-2007
Articles in Peer-reviewed Journals	Chen SS, Fitzgerald W, Zimmerberg J, Kleinman HK, Margolis L. "Cell-cell and cell-extracellular matrix interactions regulate embryonic stem cell differentiation." Stem Cells. 2007 Mar;25(3):553-61. PMID: 17332514 , Mar-2007